METRIC

A-A-54643
17 May 1992

COMMERCIAL ITEM DESCRIPTION

DEFIBRILLATOR/ECG MONITOR-RECORDER

(With External Non-Invasive Cardiac Pacing Cassette)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a defibrillator/ monitor-recorder with external non-invasive cardiac pacing cassette for defibrillation, diagnostic ECG monitoring, and non-invasive external pacing for patients in a hospital environment.

Salient characteristics:

The defibrillator/monitor-recorder shall perform defibrillation, synchronous cardioversion, heart rate detection, display and alarm, ECG waveform display and hard copy recording, and non-invasive external pacing. The system shall be integral or modular in design. It shall consist of defibrillator, pacing, and ECG monitor-recorder functional sections. Units which are modular in design and can be separated shall consist of only two separate components: a defibrillator and a monitor-recorder. Both components shall be independently powered and capable of functioning from self-contained batteries and AC line power.

The system shall be in accordance with the following national standards:

* American National Standards Institute (ANSI) and Association for the Advancement of Medical Instrumentation (AAMI) section DF 2, Standard for Cardiac Defibrillator Devices, with the exception of paragraph 3.2.1.2, Energy Range.

* Underwriters Laboratories (UL) Standard 544, Standard for Safety of Electric Medical and Dental Equipment.

AMSC/NA FSC 6515

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* National Fire Protection Association (NFPA) 99, Standard for Health Care Facilities.

* American Heart Association (AHA) publications, Recommendations for Standardization of Leads and Specifications for Instruments in Electrocardiography and Vectorcardiography.

* AAMI ECG 5/83, Standard for ECG Connectors.

The system shall be capable of full operation in the following
environmental extremes:

* Atmospheric pressure - 66 kPa - 93kPa
* Relative humidity - 95 percent (non-condensing)
* Operational temperature - 0 degrees C - 45 degrees C
* Storage temperature - -30 degrees C - 60 degrees C

The total weight of the system shall not exceed 18.1 kg.

The system shall be capable of operating over the full tolerance range of one or more of the following AC voltage and frequency combinations as specified by the customer requisitions:

* 115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
* 115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
* 230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
* 230 VAC plus or minus 10 percent, 60 plus or minus 2 Hz

Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustment to effect the conversion shall not require the use of specialized tools or skills.

The system shall also be capable of full operation from DC power derived from internal batteries. Acceptable DC battery power shall enable any of the following operational capabilities from a fully charged condition:

* A minimum of 1.25 consecutive hours of cardiac monitoring.
* A minimum of 45 minutes of combined monitoring and recording.

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* A minimum of 75 maximum energy defibrillator counter shocks into a 50 ohm resistive load at an ambient temperature of 25 degrees C.

The manufacturer shall specify a time interval (i.e. 2 years) after which the rechargeable internal batteries shall be replaced.

The system shall incorporate an integrated battery charging system which provides the following:

* Sufficient energy to fully power the unit while simultaneously charging depleted batteries.
* A charge time for fully depleted batteries to full charge not to exceed 24 hours.
* A battery level indicator which depicts the charge level of each battery used in the system. This indicator shall clearly display any low battery condition.

The system shall be provided with the following minimum accessories:

1. One (1) each three lead patient cable. Shall be shielded and have an approximate minimum length of 3 meters with lead length of approximately 1 meter and an overall cable length of approximately 4 meters.
2. Three (3) each reusable limb electrodes with straps.
3. One (1) each spare recording stylus, if applicable (w/replacement tool, if required).
4. Three (3) rolls of recording paper.
5. Three (3) sets of pre-gelled pacing electrodes.
6. Three (3) sets of disposable ECG limb electrodes.
7. One (1) each tube of electrode gel.
8. Applicable number of power cables.
9. One (1) set of external pediatric paddles.
10. One (1) each accessory bag.

The system shall be capable of displaying patient ECG signal obtained from either the defibrillator paddles or the three lead patient cable, shall incorporate a non-fade type display measuring no less than 10.1 cm diagonally, and shall provide either a freeze control or a delayed recording capability to enable capture of discrete cardiac abnormalities and arrhythmias.

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The system shall incorporate a monitor quality frequency response of no less than 0.8 Hz to 30 Hz (measured at -3 db), shall have a sweep speed of 25 mm/second, and shall incorporate an operator adjustable audible systole control.

The system shall display the external non-invasive pacing pulse on the CRT but not count the pulse on the digital heart rate display, shall enable lead selection of 3-leads (I, II, III), and shall be provided with an operator adjustable, ECG sensitivity control, to include minimum settings of 0.5., 1, and 2.

The system shall incorporate a frequency response of 0.8 Hz to 40 Hz measured at -3 db point, shall provide a common mode rejection ratio of no less than 100 db with respect to chassis ground with a 51 K Ohm plus or minus 1 percent imbalance at 60 Hz; 65 db minimum with respect to isolated ground, and shall be electrically protected from defibrillator counter shocks up to and including 360 joules (delivered).

The system shall incorporate a one millivolt reference signal.

The system shall incorporate an electrically isolated patient input circuit in accordance with the latest edition of ANSI/AAMI safe current limits for electromedical apparatus.

The recorder shall have a frequency response of 0.8 Hz to 40 Hz measured at -3 db, and shall incorporate sweep speed of 25 mm/second.

If the system is of modular design, the recorder shall be integrated with the monitor. Stand alone recorders are not acceptable.

The system shall incorporate real time recording which enables recording of events as they are being viewed on the display screen and a delay mode which enables capturing significant cardiac events after they have been viewed on the monitoring screen. Delay time shall be at least 4 seconds.

The heart rate meter shall detect R waves of either positive or negative polarity, shall incorporate a digital display which continuously indicates the patient heart rate, and shall be capable of displaying the patient heart rate within a range of 30-240 beats per minute.

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The heart rate meter shall provide updated heart rate data within a time period not to exceed five seconds.

The defibrillator shall utilize a clinically proven defibrillator pulse shape (i.e., Lown, Edmark, or Trapezoidal) as defined in AAMI Standard DF2.

The defibrillator/monitor-recorder shall operate from alternating current (AC) and direct current (DC).

It shall incorporate continuously adjustable energy settings over a range of 0-360 delivered joules presented in at least six discrete energy steps including but not limited to 10, 50, 100, 200, 300, and 360 joules, and shall charge to maximum energy selection (360 joules) using line or battery power, in a time period not to exceed 12 seconds at 28 degrees C.

The system shall incorporate a synchronization feature which synchronizes the defibrillation discharge with the patient-generated R wave; synchronization shall be visually indicated on the monitor display, marking the exact position defibrillation will be effected. The marker shall also be presented on the ECG recording.

The system shall automatically revert to the non-synchronized, defibrillation mode after each synchronized cardioversion attempt.

It shall be capable of delivering ten maximum energy discharges into a 50 ohm resistive load on battery power within five minutes at 0 degrees C.

Push button discharge controls shall be employed on both paddle handles which must be simultaneously activated for discharge to occur.

A remote button shall be located on the defibrillator chassis to initiate energy charge.

The defibrillator shall automatically discharge internally all stored energy when power to the unit is deactivated, and shall incorporate paddles which access patient ECG signal upon skin contact.

The defibrillator output pulse shall be isolated from ground when operated on AC power.

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The external, non-invasive cardiac pacing device shall enable external, non-invasive, cardiac pacing rates from 40 to 90 beats per minute, plus or minus 5 percent, and shall employ a pacing pulse which has a pulse width not greater than 40 milliseconds. The pulse width must be within 20 percent of the specified value.

The cardiac pacing device shall utilize disposable, pre-gelled electrodes, and shall be electrically protected from defibrillator counter shocks up to and including 360 joules (delivered).

If the system is modular in design, power for this device shall be delivered from the defibrillator module.

The cardiac pacing device shall provide demand and non-demand pacing modes.

The delivered output current will be adjustable over an operator selectable range of 0 to 140 milliamps in discrete steps. The accuracy of the selected current output shall be within 10 percent over the range of load impedance.

The contractor shall furnish manuals, handbooks, and/or brochures containing complete operation, installation, and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language, and all schematics and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required. All service notes, service memos, etc. pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, those furnished to service engineers (or servicemen) who normally install and service the equipment or operate ancillary equipment. The service data shall include complete instructions and drawings that show the interfacing of all system components.

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Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, only a total of two (2) service manuals need to be supplied to each consignee. However, two (2) operational manuals still need to be supplied with each unit.

Workmanship. The defibrillator/monitor-recorder shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA) One defibrillator/monitor-recorder, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one defibrillator/ monitor-recorder.

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Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one defibrillator/monitor-recorder.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Defibrillator leakage test. Chassis leakage - Leakage current measured between any exposed conductive surface and ground shall not exceed 100 uA.

Paddle leakage - Leakage current measured between an external paddle electrode and ground shall not exceed 100 uA. Leakage current measured between an internal paddle electrode and ground shall not exceed 50 uA.

Conditions - leakage current measurements will be made under all of the following conditions.

a. Power on and off.

b. Defibrillator in both charged and uncharged state.

ECG monitor/recorder leakage current test. Chassis leakage - Same as chassis leakage under defibrillator leakage test.

Lead leakage (source) - Leakage current measured between each lead and ground, all leads and ground, and between any two leads shall not exceed 10 uA.

Condition - Leakage current measurement will be performed with power on and off.

Lead leakage (sink) - Sink current measured between each lead at a potential of 120 V and ground shall not exceed 20 uA. This test shall be performed under the conditions listed above except the ground remains intact.

Strain relief test. Gradually apply a 6 N. pull in a direction that would be the most likely to pull the cord from the enclosure and maintain for one minute. The strain relief is not

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acceptable if there is such movement of the cord as to indicate that stress is being transmitted to terminals, splices or interior wiring.

Air clearance spacing test. Measurements shall be made using an appropriate measuring instrument, to verify isolated patient connected circuitry is spaced at least 3 mm inch from non-isolated circuitry.

Defibrillator paddle insulation test. Apply aluminum foil to the entire gripping surface of one paddle with sufficient pressure to make it conform to the surface.

With the paddles connected to the defibrillator, connect the high potential lead of a 60 Hz (or 50 Hz, if preferred) high potential test source to the aluminum foil. Connect low-potential or ground lead to the paddle electrode. Slowly adjust the peak voltage amplitude of the test source to equal 1.5 times the maximum voltage occurring on the defibrillator energy stored capacitor. Maintain the voltage for 1 minute.

Repeat this test with the low-potential lead connected to the other paddle electrode.

There shall be no voltage breakdown, and the 60 Hz leakage current shall not exceed 250 uA peak (177 uA rms); if a 50 Hz test source is used, the current shall not exceed 210 uA peak (148 uA rms).

ECG monitor/recorder sweep/chart speed test. Measure the monitor sweep speed and the recorder paper speed by applying a signal of known frequency to the patient cable. Monitor and recorder chart sweep speed shall be within plus or minus 5 percent of 25 mm/sec.

ECG amplifier calibration test. The accuracy of the 1 mV internal calibration pulse shall be measured by comparing the amplitude of this pulse to a pulse applied to the patient cable from a pulse generator. It shall be accurate to plus or minus 15 percent.

ECG amplifier CMRR test. Common mode rejection is measured by comparing the amplitude of a 1 mV differential signal applied to the monitor to the amplitude of a higher voltage sinusoidal signal of 60 Hz applied between the junction of the leads and ground. Measured common mode rejection shall be at least 100 db with respect to chassis ground with 5 k ohms imbalance between leads.

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Defibrillator energy accuracy test. Connect the defibrillator paddles to a defibrillator analyzer equipped with a 50 ohm resistive load. Charge the defibrillator to a pre-selected energy level, then discharge into the analyzer. Repeat for each available energy level setting. Measure and record data. Measured energy must be within plus or minus 15 percent (or 4 joules, whichever is greater) of the selected energy.

Defibrillator charge time test. Using a stopwatch, measure the time between activation of the charge control and full charge indication for the first and last of 10 sequential charges and discharges all at 320 joules or above. This time shall not exceed 12 seconds.

Synchronized discharge test. Connect a signal generator to the ECG monitor. Measure the time between the leading edge of square pulse and the leading edge of the defibrillator's synchronized discharge pulse. This time can be displayed automatically on a defibrillator analyzer if so equipped, or measured directly from an oscilloscope, and shall be within 30 milliseconds.

ECG monitor and recorder frequency response test. Apply a 1 mV peak to peak differential sinusoidal signal to the patient cable, and vary the frequency. The bandwidth is defined by the upper and lower frequencies at which the trace deflection amplitude decreases to 0.707 times the maximum value. This bandwidth shall be 1.0 Hz to 30 Hz on the cardioscope and on the recorder.

Humidity test. Place the unit in test chamber such that leakage current and delivered energy accuracy test can be performed external to the chamber.

Adjust the temperature of the chamber to 25 degrees C.

Permit sufficient time for the temperature "inside the unit" to stabilize prior to proceeding.

After temperature stabilization, adjust the relative humidity to 95 percent and permit sufficient time for humidity stabilization.

After 48 hours of conditioning, source and sink leakage current test are to be performed. Record results on leakage current summary charts.

Perform the energy accuracy test as specified above (Defibrillator Energy Accuracy Test).

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The unit shall be in compliance, if leakage current and energy accuracy are within the specified limits.

Battery capacity test. Verify 2.25 hours of continuous cardioscope monitoring or 45 minutes of continuous recording of a heart rate of 60 BPM. Defibrillator battery will be tested by verifying twenty-five maximum 320 joules discharges at 25 degrees C.

Defibrillator discharge pulse rate test. Apply the paddle electrodes to a defibrillator tester and select the maximum energy level. Simultaneously initiate the charge and a timer.

Immediately upon completion of the charge, discharge the defibrillator.

The defibrillator shall be capable of delivering 15 maximum energy pulses into a 50 ohm resistive load within 5 minutes, at 0oC.

Defibrillator overload protection test. Overload protection shall comply with those requirements specified in ANSI/AAMI EC 13-1983, American National Standard for Cardiac Monitors, Heart Rate Meters and Alarms.

Electrical characteristics test. Operate monitor/recorder and defibrillator units over the full tolerance range of the following line voltage and frequency combinations:

115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
230 VAC plus or minus 10 percent, 60 plus or minus 2 Hz

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/ suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA - MPQAS
USPHS
VA - OSS

Project Number: 6515-5337

Location: DEFC17B.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54643	2. DOCUMENT DATE (YYMMDD) 17 May 1992

3. DOCUMENT TITLE: DEFIBRILLATOR/ECG MONITOR-RECORDER (With External Non-Invasive Cardiac Pacing Cassette)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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